

Business Insights & Recommendations

Zomato Business Intelligence & Delivery Time Prediction Platform

For: City Operations, Restaurant Partnerships, Growth Marketing, Finance | Author: Sundram Sharma | Evidence: cleaned data (24 months), 23 permutation tests, 2 model families

1. Executive summary

The problem to solve is order failure, not delivery speed. 33.9% of orders never reach the customer (22.3% cancelled, 11.5% 'Food Not Delivered'), and 15.9% of the rest arrive late. Revenue is INR 53.1 lakh over 24 months and flat month to month.

Nothing we measured explains the failures or the delays. City, cuisine, weather, traffic, payment method, vehicle, hour, weekend and membership show no statistically reliable effect (only 2 of 23 tests at $p < 0.05$, about what chance produces). Consequently **neither predictive model beats a trivial baseline**: the delivery-time models reach test $R^2 = 0.001$ (PRD target 0.75) and the churn models reach ROC-AUC 0.53 (0.50 = coin flip).

Recommendation: do not deploy either model and do not reallocate partners or target campaigns using them. Spend the next sprint on instrumentation (cancellation reasons, stage timestamps, GPS, customer coordinates), and on three decisions that do not depend on the models: reconcile payments on failed orders, run a coupon holdout test, and use the restaurant scorecard only as a watch-list.

2. What the data does show

Area	Finding	Why it matters
Order failure	33.9% of orders (4,569 cancelled, 2,362 not delivered)	Largest controllable loss
Late deliveries	15.9% of completed; late orders average 63.7 min vs 33.5	SLA breach
Delivery time	mean 38.3, p90 60, p95 70 min	Baseline for SLA
Peak demand	12-13h and 19-20h daily; weekends not higher	Only strongly structured pattern
Customers	37.8% of ordering customers bought once; 18.1% of registrants never ordered	Activation gap
Payments	7.9% of orders have a Failed payment	Reconciliation
Coupons	35% of orders use a coupon, avg discount 28% of food cost; INR 5.1 lakh discounted on completed orders	Cost with unknown benefit

3. What the data does not show (and why that matters)

Each claim below was tested with a permutation test (2,000 shuffles). A pattern with $p \geq 0.05$ is not evidence of an effect, and with 23 tests one 'hit' is expected by chance.

Hypothesis	Observed gap	p-value	Reading
Delivery time vs traffic level	0.4 min longer in High/Severe	0.36	No effect detected
Delivery time vs rain	0.1 min longer in heavy rain	0.47	No effect detected
Late rate vs peak hour	-	0.85	No effect detected
Cancellation vs cuisine	19.0% (Chinese) to 24.9% (Mexican)	0.058	Borderline, 1 of 23 tests
Cancellation vs payment method	20.9% (COD) to 23.8% (Net Banking)	0.036	Weak; expected by chance
Cancellation vs weather / city / hour	-	0.83 / 0.95 / 0.84	No effect detected
Basket vs membership tier	INR 389 - 395	0.84	No effect detected
Partner speed differences	287 partners with 10+ deliveries	0.09	Mostly noise

4. Model results

Delivery time (regression, 2,691 held-out completed orders). Selected by cross-validation: Random Forest (tuned)

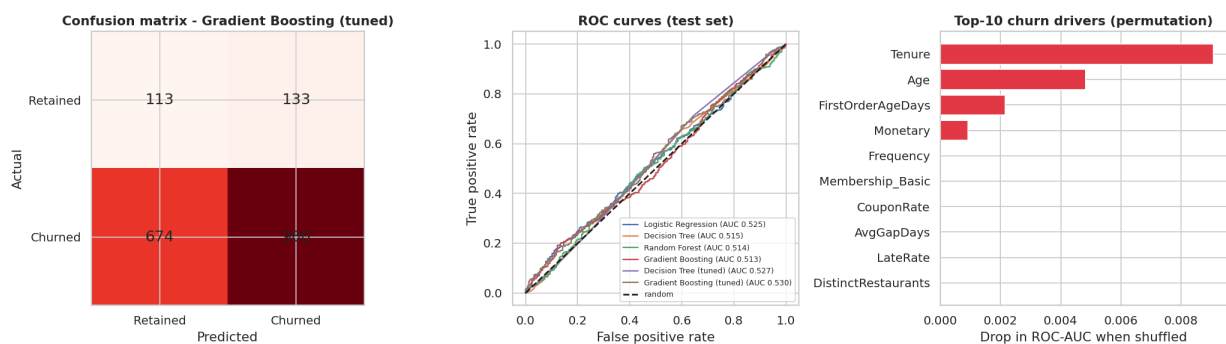
Delivery-time model	CV R2	MAE	RMSE	Test R2
Baseline (predict train mean)	0.0	11.524	15.138	-0.0
Linear Regression	-0.005	11.542	15.154	-0.002
Decision Tree	-0.033	11.528	15.179	-0.005
Random Forest	-0.007	11.505	15.142	-0.001
Gradient Boosting	-0.009	11.523	15.158	-0.003
Random Forest (tuned)	-0.002	11.501	15.132	0.001
Gradient Boosting (tuned)	-0.003	11.514	15.133	0.001
LEAKAGE DEMO: GB + OrderStatus flag (not usable)	-	8.313	10.289	0.538

No model improves on predicting the average (15.1 min RMSE). The row marked LEAKAGE adds the post-delivery late flag and reaches $R^2 = 0.54$: it demonstrates why OrderStatus is excluded, not a usable model. Reported feature 'importances' sit at the noise floor and should not be read as drivers.

60-day churn (classification, 9,498 customers; 87% churn). Selected by CV AUC: Gradient Boosting (tuned)

Churn model	CV AUC	Accuracy	F1 (churn)	ROC-AUC	PR-AUC
Baseline: predict everyone churns	0.5	0.871	0.931	0.5	0.871
Logistic Regression	0.5	0.528	0.663	0.525	0.885
Decision Tree	0.506	0.346	0.437	0.515	0.872
Random Forest	0.495	0.761	0.861	0.514	0.875
Gradient Boosting	0.499	0.576	0.713	0.513	0.88
Decision Tree (tuned)	0.503	0.667	0.789	0.527	0.882
Gradient Boosting (tuned)	0.519	0.575	0.708	0.53	0.887

The 'everyone churns' rule scores $F1 = 0.93$; every real model scores lower, and ROC-AUC is 0.51-0.53. The PRD's $F1 \geq 0.70$ target is therefore not meaningful here. On held-out customers the High/Medium/Low risk segments churn at 88.1% / 88.0% / 85.0% - a few points apart (in-sample the gap looks larger, 91% vs 83%, which is overfitting) - so they cannot support targeting.



5. Recommendations by stakeholder

City Operations - fix visibility before reallocating partners

- **Why:** 33.9% of orders fail and 15.9% of completed ones are late, yet no measured factor explains either, so reallocating partners by traffic/weather/hour would act on noise.
- **Do:** log a cancellation/failure reason code and the actor (customer, restaurant, partner, system) plus timestamps for accepted, prepared, picked-up, delivered. Add partner GPS and customer coordinates for true distance.
- **Measure success:** share of failures with a reason code (target 95%); then the share of failures attributable to controllable causes. Re-run the delivery-time model after 4 weeks of new data and require R^2 on a held-out time period well above 0 before acting on importances.

Restaurant Partnerships - scorecard as a watch-list only

- The composite score (0.4 rating, -0.3 cancel rate, -0.3 late rate; *restaurant_scorecard.csv*) ranks 1,200 restaurants, but the median restaurant has only 17 orders (max 35), so its cancel rate has a standard error near 10 percentage points.
- **Do:** review only restaurants that stay in the bottom decile for three consecutive months with at least 30 orders; do not use rating as a quality proxy - it is uncorrelated with cancellations ($r = 0.01$).

Growth Marketing - test coupons, stop relying on the churn score

- 35% of orders carry a coupon averaging 28% off, costing INR 5.1 lakh on completed orders versus INR 53.1 lakh revenue. Observational data cannot show whether coupons create orders or just discount orders that would have happened.
- **Do:** run a randomised coupon holdout (e.g. 10% of eligible users get no coupon for 4 weeks) and compare orders per user and margin. For retention, use the simple rule 'no order in 60 days' - it performs as well as the model - and activate the 18% of registrants who never ordered (first-order offer).

Finance - payment reconciliation and a flat forecast

- Among 6,931 cancelled / not-delivered orders, 5,440 (INR 21.3 lakh) show payment 'Success' and only 7.8% show 'Refunded', the same mix as delivered orders. Either status fields are unsynchronised or customers may be owed refunds.
- **Do:** reconcile payment gateway records against order status. For planning, monthly revenue is stable (mean INR 2.21 lakh, SD 0.12); a naive next-quarter estimate is about INR 6.7 lakh (trailing 3-month average $\times 3$), with no evidence of growth or seasonality in the 24 months available.

6. Risks and limits of this analysis

- Revenue counts completed orders only; ~34% of orders produced nothing, and payment data suggests some may still have been charged.
- Ambiguous slash dates (~15%) were resolved heuristically; weather/traffic link to only ~63% of orders. Neither changes the conclusions (all weather/traffic effects are null on the linked subset).
- The data behaves like a simulated snapshot: many natural relationships (lateness vs rating, RFM vs churn, traffic vs time) are absent. Conclusions apply to this dataset; validate on production data before operational use.
- Model numbers in this report were produced with the project's NumPy fallback implementation of scikit-learn; re-running *python run_pipeline.py* with scikit-learn installed regenerates every figure, table and model file.